CIRC HQ STUDENT HANDOUT | GRADE 5

Binary Beacons

Goal: Encode and decode a message using a documented two-state binary system.

35 minutes | 8 steps | Computer Science and Programming | Playbook B meeting 16, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Provide the key, encoded message, decoded result, and one corrected transmission error.

You will use

- Two-color cards
- Paper cups
- Flashlights used at tables
- Binary place-value mats
- Message slips

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Choose two state cards.
- Place the beacon at table level.

2 Step 2: Light safety — 3 min

- Keep every light pointed at the table.
- Turn off any light aimed near a face.
- Use only classroom-safe messages.

3 Step 3: Build the key — 5 min

- Assign a value to each state.
- Write the value in each number spot.
- Test one sample number.

4 Step 4: Encode a message — 7 min

- Choose a short classroom message.
- Convert each symbol with the key.
- Place the states in order.

5 Step 5: Send the states — 6 min

- Show one fixed position at a time.
- Pause between symbols.
- Let the partner record silently.

6 Step 6: Check the message — 5 min

- Compare the partner answer with the original message.
- Correct one unclear state.

7 Step 7: Power down — 3 min

- Turn off every light.
- Stack the state cards.

8 Step 8: Exit check — 3 min

- Write one rule that prevented confusion.

What to turn in

Your key, coded message and decoded result.

No computer? Another way to do it

Use cards only, no flashlights.

Finished early? Next challenge

Error Check: Add one check bit or repeated pattern that helps the receiver notice a transmission error.

Stay safe

Do not shine lights toward faces and use only classroom-safe, nonpersonal messages.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.